LAMBDA and related Func

RESULT	TRUE	
Sheet	**Result**	**Comment**
Function_LAMBDA	TRUE	
Function_ISOMITTED	TRUE	
Function_BYCOL	TRUE	
Function_BYROW	TRUE	
Function_MAKEARRAY	TRUE	
Function_MAP	TRUE	
Function_REDUCE	TRUE	
Function_SCAN	TRUE	

ctions

FUNCTION	EXPECTED	CORRECT
2	2	TRUE
10	10	TRUE
3	3	TRUE
4	4	TRUE
7	7	TRUE
10	10	TRUE
15	15	TRUE
5	5	TRUE
hi!	hi!	TRUE
2	2	TRUE
100	100	TRUE
6	6	TRUE
0	0	TRUE
TRUE	TRUE	TRUE
2	2	TRUE
3	3	TRUE
7	7	TRUE
16	16	TRUE
6	6	TRUE
82	82	TRUE
200	200	TRUE
200	200	TRUE
120	120	TRUE
7	7	TRUE
4	4	TRUE
100	100	TRUE
110	110	TRUE
3	3	TRUE
4	4	TRUE
10	10	TRUE
7	7	TRUE
32	32	TRUE
15	15	TRUE
333	333	TRUE
20	20	TRUE
2	2	TRUE
7	7	TRUE
xy	xy	TRUE
Syntax Error	Manual check	
Syntax Error	Manual check	
Syntax Error	Manual check	
Syntax Error	Manual check	
Syntax Error	Manual check	
Syntax Error	Manual check	
Syntax Error	Manual check	
Syntax Error	Manual check	
Syntax Error	Manual check	

FORMULA

```
=LAMBDA(a, 2)(1)
=LAMBDA(a, a)(F2)
=LAMBDA(a, a)(3)
=LAMBDA(a, a * 2)(2)
=LAMBDA(x,y, x + y)(3, 4)
=LAMBDA(a, a + a)(5)
=LAMBDA(x,y, x * y + x)(3, 4)
=LAMBDA(a, IF(a > 0, a, -a))(-5)
=LAMBDA(a, a & "!")("hi")
=LAMBDA(SQRT, SQRT(SQRT))(4)
=LAMBDA(r, SUM(r))(F2:F5)
=LAMBDA(a,b,c, a + b + c)(1, 2, 3)
=LAMBDA(a,b,c, a + b + c)(,,)
=LAMBDA(x, x > 5)(7)
=ERROR.TYPE(LAMBDA(x, 1 / x)(0))
=ERROR.TYPE(LAMBDA(x,y, x + y)(1))
=LAMBDA(x, LAMBDA(y, x + y))(3)(4)
=LET(f, LAMBDA(x, x * x), f(4))
=LAMBDA(a, SUM(a))({1;2;3})
=SUM(LAMBDA(a, a * 2)({10;11;20}))
=SUM(LAMBDA(a, a * 2)(F2:F5))
=SUM(LAMBDA(a, a * 2)(DATARANGE))
=LET(g, LAMBDA(self,n, IF(n <= 1, 1, n * self(self, n - 1))), g(g, 5))
=PLUS2(5)
=SUM(4)
=LAMBDA(x, SUM(x))(100)
=LAMBDA(x,f, f(x))(100,SUM)
=LAMBDA(f,n, f(n))(PLUS2, 1)
=LAMBDA(f,n, f(n))(COUNT, {1,2,3,4})
=LAMBDA(f, f(5))(LAMBDA(x, x * 2))
=LAMBDA(f,x, f(f(x)))(LAMBDA(n, n + 1), 5)
=APPLYTWICE(LAMBDA(x, x * 2), 8)
=LAMBDA(x, x + $F$2)(5)
=LAMBDA(a, LAMBDA(b, LAMBDA(c, a + b + c)))(3)(30)(300)
=LAMBDA(x,y, x + y)(20,)
=LAMBDA(x,y, x + y)(,2)
=LAMBDA(x,[y],[z], x + y + z)(7)
=LAMBDA(x,[y],[z], x & y & z)("xy")
=LAMBDA(AX1, AX1, 0)
=LAMBDA([x], y, 0)
=LAMBDA(x$, 0)
=LAMBDA(ab!, 0)
=LAMBDA(aa#a, 0)
=LAMBDA(a%, 0)
=LAMBDA(xx@3, 0)
=LAMBDA(x,y,z, {x,y,z}))(1,2,3)
=LAMBDA("x", 0)(1)
```

COMMENT
Constant
Reference as input
Value as input

Unspecified value is defaulting to 0

Div by 0
Missing required parameter

LAMBDA in a LET function

Array input
Range input
Name (of a range) input
Recursive LAMBDA
PLUS2 set with LAMBDA in the Name Manager
SUM LAMBDA function is set in Name Manager, make sure built-in function has priority
If you use SUM in LAMBDA, it will use the built-in function not the named LAMBDA
If you use SUM as parameter to LAMBDA, the named LAMBDA will be used instead
Named lambda as parameter
ETA-LAMBDA use in LAMBDA as parameter
LAMBDA as parameter to LAMBDA
Call LAMBDA 2 times
Export of a named LAMBDA that takes a LAMBDA argument
A LAMBDA body that closes over a sheet cell
Tripple curry
Value not written - second parameter
Value not written - first parameter
TODO - Optional used but not set - number
TODO - Optional used but not set - string
AX1 = cell reference name - invalid syntax
Optional, then required parameter - invalid syntax

Can't create an array from parameters

Data

10
20
30
40

FUNCTION	EXPECTED	CORRECT
TRUE	TRUE	TRUE
FALSE	FALSE	TRUE
FALSE	FALSE	TRUE
104	104	TRUE
8	8	TRUE
3	3	TRUE
-1	-1	TRUE
5	5	TRUE
12	12	TRUE
3	3	TRUE

FORMULA

```
=LAMBDA(x,y, ISOMITTED(x))(, 5)
=LAMBDA(x,y, ISOMITTED(x))(3, 5)
=LAMBDA(x,y, ISOMITTED(x))(3, 5)
=LAMBDA(x,y, IF(ISOMITTED(x), 99, x) + y)(, 5)
=LAMBDA(x,y, IF(ISOMITTED(x), 99, x) + y)(3, 5)
=LAMBDA(x,y, ISOMITTED(y))(5)
=LAMBDA(a,b,c, IF(ISOMITTED(b), -1, b))(1, , 3)
=LAMBDA([x], IF(ISOMITTED(x),12,5))(1)
=LAMBDA([x], IF(ISOMITTED(x),12,5))()
=LAMBDA(x, IF(ISOMITTED(x),12,5))()
```

COMMENT

#VALUE!

TODO - Optional parameter, set
TODO - Optional parameter, not set
TODO - Required parameter, not set #VALUE!

FUNCTION	EXPECTED	CORRECT
21	21	TRUE
6	6	TRUE
11	11	TRUE
22	22	TRUE
33	33	TRUE
100	100	TRUE
ABC	ABC	TRUE
AD	AD	TRUE
BE	BE	TRUE
CF	CF	TRUE
132	132	TRUE
165	165	TRUE
198	198	TRUE

FORMULA

```
=SUM(BYCOL({1,2,3;4,5,6}, LAMBDA(c, SUM(c))))
=BYCOL({1;2;3}, LAMBDA(c, SUM(c)))
=TRANSPOSE(BYCOL({1,2,3;10,20,30}, LAMBDA(c, SUM(c))))


=BYCOL(100, LAMBDA(c, SUM(c)))
=BYCOL("ABC", LAMBDA(c, c))
=TRANSPOSE(BYCOL({"A","B","C";"D","E","F"}, LAMBDA(c, CONCAT(c))))


=BYCOL(F2:H4, LAMBDA(c, SUM(c)))
```

COMMENT	Data		
	11	22	33
	44	55	66
	77	88	99
Column reduction over a real range	132	165	198

FUNCTION	EXPECTED	CORRECT
21	21	TRUE
6	6	TRUE
6	6	TRUE
60	60	TRUE
100	100	TRUE
ABC	ABC	TRUE
ABC	ABC	TRUE
DEF	DEF	TRUE
66	66	TRUE
165	165	TRUE
264	264	TRUE
3	3	TRUE
7	7	TRUE

FORMULA

```
=SUM(BYROW({1,2,3;4,5,6}, LAMBDA(c, SUM(c))))
=BYROW({1,2,3}, LAMBDA(c, SUM(c)))
=BYROW({1,2,3;10,20,30}, LAMBDA(c, SUM(c)))

=BYROW(100, LAMBDA(c, SUM(c)))
=BYROW("ABC", LAMBDA(c, c))
=BYROW({"A","B","C";"D","E","F"}, LAMBDA(c, CONCAT(c)))

=BYROW(F2:H4, LAMBDA(c, SUM(c)))


=BYROW({1,2;3,4}, LAMBDA(r, REDUCE(0, r, LAMBDA(a,v, a + v))))
```

COMMENT **Data**

11	22	33
44	55	66
77	88	99

Row reduction over a real range

FUNCTION	EXPECTED	CORRECT
11	11	TRUE
12	12	TRUE
13	13	TRUE
21	21	TRUE
22	22	TRUE
23	23	TRUE
2	2	TRUE
3	3	TRUE
2	2	TRUE
3	3	TRUE
6	6	TRUE
6	6	TRUE

FORMULA	COMMENT
=TRANSPOSE(MAKEARRAY(2, 3, LAMBDA(r,c, r * 10 + c)))	
=MAKEARRAY(1, 1, LAMBDA(r,c, r + c))	
=MAKEARRAY(0, 3, LAMBDA(r,c, 1))	
=MAKEARRAY(2, 1, LAMBDA(r,c, r + c))	
=MAKEARRAY(2, 1, LAMBDA(r,c, 6))	

DATA

11	21
12	22
13	23

#VALUE!

FUNCTION	EXPECTED	CORRECT
1	1	TRUE
4	4	TRUE
9	9	TRUE
11	11	TRUE
22	22	TRUE
33	33	TRUE
A	A	TRUE
B	B	TRUE
4	4	TRUE
25	25	TRUE
4	4	TRUE
1	1	TRUE
2	2	TRUE
4	4	TRUE
6	6	TRUE
1.5	1.5	TRUE
2	2	TRUE
4	4	TRUE
7	7	TRUE

FORMULA

```
=MAP({1;2;3}, LAMBDA(x, x * x))


=MAP({1;2;3}, {10;20;30}, LAMBDA(a,b, a + b))


=MAP({"a";"b"}, LAMBDA(x, UPPER(x)))

=MAP(F2:F5, LAMBDA(x, x * x))



=LET(double, LAMBDA(x, x * 2), MAP({1;2;3}, double))


=SUM(IFERROR(MAP({1,0,2}, LAMBDA(x, 1 / x)), 0))
=MAP({1;2;3}, {1;2}, LAMBDA(a,b, a + b))
```

COMMENT Data

2
5
2
1

Element map over a range

2
4
#N/A

FUNCTION	EXPECTED	CORRECT
10	10	TRUE
24	24	TRUE
abc	abc	TRUE
10_abc	10_abc	TRUE
6	6	TRUE
26	26	TRUE
39	39	TRUE
18	18	TRUE
-1	-1	TRUE

FORMULA

```
=REDUCE(0, {1,2,3,4}, LAMBDA(acc,val, acc + val))
=REDUCE(1, {1,2,3,4}, LAMBDA(acc,val, acc * val))
=REDUCE("", {"a","b","c"}, LAMBDA(acc,val, acc & val))
=REDUCE(10 & "_", {"a","b","c"}, LAMBDA(acc,val, acc & val))
=REDUCE(, {1,2,3}, LAMBDA(acc,val, acc + val))
=REDUCE(5, {1,2,3;4,5,6}, LAMBDA(acc,val, acc + val))
=REDUCE(2, F2:F4, LAMBDA(acc,val, acc + val))
=REDUCE(10, UNIQUE(G2:G4), LAMBDA(acc,val, acc + val))
=IFERROR(REDUCE(0, {1,0}, LAMBDA(acc,val, acc / val)), -1)
```

COMMENT	Data 1	Data 2
	12	3
	14	3
	11	5

REDUCE over a 2D array
REDUCE over a range

FUNCTION	EXPECTED	CORRECT
1	1	TRUE
3	3	TRUE
6	6	TRUE
10	10	TRUE
1	1	TRUE
3	3	TRUE
6	6	TRUE
aa	aa	TRUE
aab	aab	TRUE
aabc	aabc	TRUE

FORMULA

```
=TRANSPOSE(SCAN(0, {1,2,3,4}, LAMBDA(acc,val, acc + val)))
```

```
=TRANSPOSE(SCAN(, {1,2,3}, LAMBDA(acc,val, acc + val)))
```

```
=SCAN("a", {"a";"b";"c"}, LAMBDA(acc,val, acc & val))
```

COMMENT